

Uncertainty-Aware Active Multi-View Pose Estimation for Robotic Arms

Wendi Gan, Wei Zhai*, Yang Cao, and Zheng-Jun Zha

MoE Key Laboratory of Brain-inspired Intelligent Perception and Cognition,
University of Science and Technology of China
gwdx@mail.ustc.edu.cn {wzhai056,forrest,zhazj}@ustc.edu.cn

Abstract. Accurate pose estimation is a fundamental requirement for high-precision mobile manipulation. However, the conventional single-viewpoint localization method is highly susceptible to viewpoint degeneracy and environmental noise, while existing multi-view fusion strategies determine per-view contributions through idealized noise assumptions. To address these limitations, we propose an uncertainty-aware active multi-view pose estimation framework. The framework decouples uncertainty estimation from pose fusion into two stages. The offline stage estimates viewpoint-specific error covariances in the Lie algebra space through repeated sampling. The online stage reuses these covariances to perform quality-aware weighted fusion in real time, suppressing spatially degraded viewpoints. Experimental results demonstrate that the proposed method consistently reduces both translational and angular errors compared to reprojection-based fusion. Real-world validations on a physical mobile manipulator confirm sub-millimeter repeatability under full operational conditions, successfully enabling precise manipulation tasks in a chemical laboratory.

Keywords: Visual pose estimation · Robotic arm · Multi-view fusion
· Uncertainty-aware weighting

1 Introduction

Accurate pose estimation is essential for high-precision robotic manipulation, enabling robots to localize objects, tools, and workstations for tasks such as pick-and-place, peg-in-hole assembly, and visual servoing. Among existing solutions, vision-based fiducial localization has become a widely adopted approach [9, 15], recovering the 6-DoF tag pose from single or multiple camera viewpoints. This capability is particularly valuable in self-driving laboratories (SDLs), an emerging paradigm where mobile manipulators autonomously navigate among experimental stations to execute complex procedures [1]. In such environments, reliable fiducial-based pose estimation supports station localization, tool alignment, and precise instrument interaction. Recent SDL platforms for autonomous chemistry and biology [3–5, 29] require robots to repeatedly transition between workstations and interact with delicate labware under tight spatial tolerances. Consequently,

achieving sub-millimeter pose estimation has become a key enabler for safe and robust SDL automation.

However, single-viewpoint observations typically suffer from accuracy degradation caused by viewpoint degeneracy and environmental noise, often failing to meet sub-millimeter precision requirements. While multi-view fusion improves accuracy through complementary geometric constraints, existing methods rely on idealized assumptions that limit their effectiveness. Equal-weight averaging [11] fuses poses uniformly, ignoring viewpoint-specific quality differences; minimizing reprojection error [2] implicitly weights views through keypoint-level optimization, but assumes identical noise distributions across all viewpoints—an assumption that rarely holds under varying observation geometries.

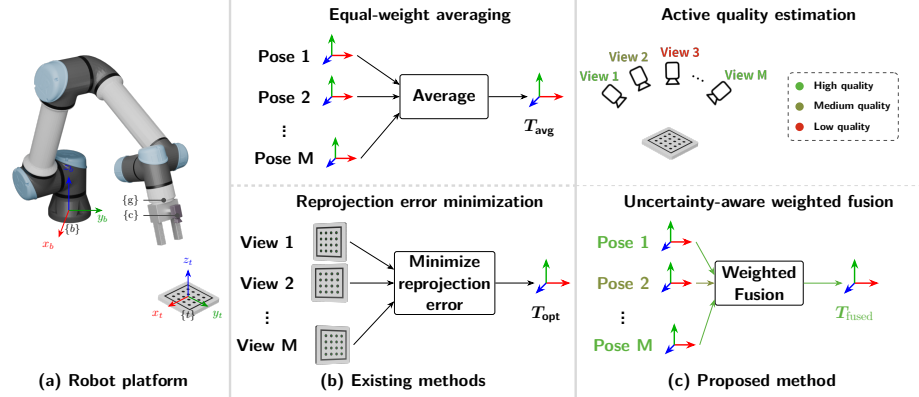


Fig. 1: Overview of multi-view pose estimation. (a) Mobile manipulator with coordinate frames. (b) Two existing fusion strategies: equal-weight averaging (top) and reprojection error minimization (bottom). (c) Proposed method: active per-viewpoint quality estimation via repeated sampling (top), and uncertainty-aware weighted fusion using the learned quality priors (bottom).

To address these limitations, this paper proposes an uncertainty-aware, weighted multi-view pose fusion method based on uncertainty modeling (Fig. 1). Rather than relying on idealized noise assumptions, our method actively maneuvers the robotic arm to perform repeated multi-view sampling during an initialization stage, empirically estimating the pose error characteristics of each viewpoint through statistical analysis. The resulting viewpoint-specific information matrix weights are then used for adaptive fusion in the Lie algebra $\mathfrak{se}(3)$ tangent space, whose linear structure allows small pose deviations to be combined with heterogeneous per-viewpoint weights.

We evaluate the proposed method on a physical mobile manipulation platform through multi-level experiments. First, benchmark comparisons demonstrate that our method consistently outperforms conventional equal-weight and reprojection-based fusion strategies. Second, real-world validation via dial indi-

cator measurements confirms sub-millimeter positioning repeatability under full operational conditions including navigation deviations. Finally, a glassy carbon plate placement task further verifies that the gained repeatability translates into reliable autonomous manipulation in a chemical laboratory setting.

The main contributions are:

- We propose an uncertainty-aware active multi-view pose estimation framework that explicitly addresses the precision degradation caused by imbalanced viewpoint quality, enabling sub-millimeter robotic arm localization.
- We introduce an uncertainty modeling approach that actively estimates viewpoint-specific residual covariances in $\mathfrak{sc}(3)$ via repeated sampling, and constructs information matrices for adaptive weighted fusion, thereby enabling quality-driven unequal-weight aggregation of multi-view poses.
- We validate the proposed method on a physical mobile manipulation platform, where real-world tests confirm reduced angular error, sub-millimeter repeatability, and reliable chemical laboratory operation.

2 Related Work

2.1 Single-View Pose Estimation

Recent advances in generative modeling and large-scale visual pre-training [8, 23, 25–27] have significantly boosted performance across a wide range of visual tasks, including 6-DoF pose estimation [13, 21, 22]. However, these methods typically operate at millimeter-level precision and require large per-configuration training datasets, making them ill-suited for the sub-millimeter, calibration-centric setting targeted here. Consequently, structured fiducial markers such as ArUco [9] and AprilTag [15] remain the standard for high-precision robotics, enabling fast and robust localization via Perspective-n-Point (PnP) solvers.

Nevertheless, single-view marker pose estimation is highly sensitive to viewing geometry. Studies [10, 16, 18] show that viewpoint degeneracy—especially when the camera optical axis becomes nearly parallel to the marker plane—can significantly amplify rotational errors, and that optimal accuracy typically requires a moderate inclination angle (e.g., 40° – 70°). These findings confirm that single-view poses exhibit strong viewpoint-dependent heteroscedasticity, which must be considered when aggregating multiple observations.

2.2 Multi-View Pose Estimation

To improve robustness against degraded single-view observations, multi-view approaches introduce complementary geometric constraints. While dense fusion methods (e.g., point clouds or high-level features) [7, 12] are effective for complex objects, they are computationally heavy for structured planar markers.

For sparse keypoints, a mainstream paradigm formulates multi-view estimation as a non-linear optimization problem that minimizes the summed reprojection error across views [2, 6, 24]. These methods exploit multi-view redundancy

but assume homoscedastic pixel-wise residuals—an assumption that rarely holds under varying observation geometries, limiting overall accuracy.

Alternatively, pose-level fusion directly aggregates multiple estimated poses [11]. Equal-weight averaging implicitly assumes identical per-view uncertainty and therefore cannot suppress geometrically degenerate viewpoints.

2.3 Uncertainty-Aware Weighting for Pose Fusion

Accurate uncertainty modeling is essential for fusing heteroscedastic observations. In foundational vision tasks, uncertainty-aware mechanisms have been incorporated to better handle heterogeneous noise, such as explicit covariance estimation in PnP solvers [28] and uncertainty weighting in calibration frameworks [19, 20]. Popescu et al. [17] adopt an adaptive Kalman filter for multi-camera localization, updating measurement covariances online as observations accumulate. In contrast, our method uses a single eye-in-hand camera that visits planned viewpoints sequentially; since the robot is stationary during each localization cycle, fusion reduces to a static problem with no state propagation. We therefore calibrate offline: repeated sampling during initialization yields fixed per-viewpoint information matrices, reused as priors thereafter.

Rather than assuming a shared parametric noise model across viewpoints, our method empirically characterizes viewpoint-specific pose residual covariances in the Lie algebra space via repeated statistical sampling. The resulting per-viewpoint information matrices enable adaptive, quality-aware pose-level fusion that suppresses geometrically degraded viewpoints while emphasizing reliable observations.

3 Method

3.1 Task Definition

The core task is to estimate a highly repeatable relative pose between the mobile robot and the workspace. In teach-and-replay paradigms, static systematic biases (e.g., from kinematics or calibration) are implicitly absorbed into the relative taught poses. Therefore, the primary challenge upon redocking is obtaining highly repeatable pose estimates to reliably reconstruct these taught tasks, making repeatability the most critical metric for operational success.

We consider an eye-in-hand configuration where the camera is rigidly mounted on the robot’s end-effector. We define four primary coordinate systems (Fig. 1): the robot base frame $\{b\}$, the end-effector frame $\{g\}$, the camera frame $\{c\}$, and the fiducial tag frame $\{t\}$. For a single observation, the pose of the tag relative to the base frame, denoted as ${}^b_t\mathbf{T} \in \text{SE}(3)$, is determined by the kinematic chain:

$${}^b_t\mathbf{T} = {}^b_g\mathbf{T} \cdot {}^g_c\mathbf{T} \cdot {}^c_t\mathbf{T}, \quad (1)$$

where ${}^b_g\mathbf{T}$ is the end-effector pose derived from the robot’s forward kinematics, ${}^g_c\mathbf{T}$ is the hand-eye transformation obtained from offline calibration, and ${}^c_t\mathbf{T}$ is the visual pose estimated by Perspective-n-Point (PnP) from the image.

However, a single-view visual estimate ${}^c_t\mathbf{T}$ is often corrupted by viewpoint-dependent degeneracy and image noise. Therefore, our multi-view localization task aims to aggregate M distinct observations $\left\{({}^b_g\mathbf{T}_m, {}^c_t\mathbf{T}_m)\right\}_{m=1}^M$ obtained from specially planned viewpoints, solving for an optimal fused pose ${}^b_t\mathbf{T}_{\text{fused}}$ that minimizes the influence of these degradation factors.

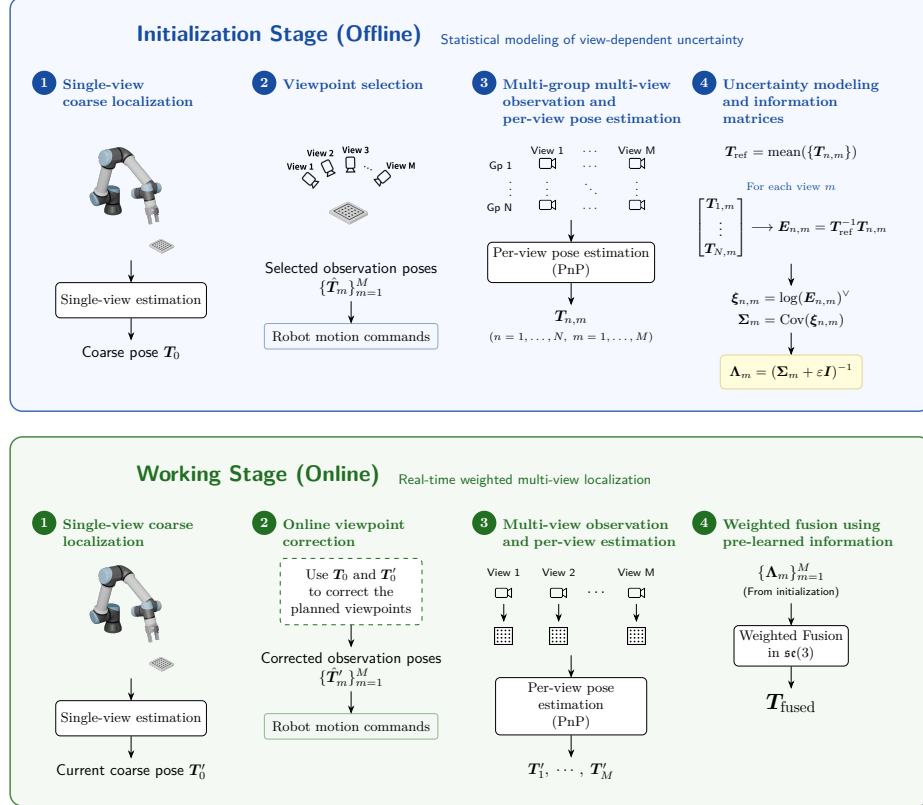


Fig. 2: Overall framework of the proposed multi-view localization method. The offline initialization stage performs repeated multi-view sampling to compute per-viewpoint information matrices $\mathbf{A}_m$; the online working stage reuses them for real-time weighted fusion.

3.2 Proposed Framework

Due to the coupled effects of robotic arm kinematic errors and viewpoint-dependent visual detection noise, pose estimates at different viewpoints follow distinct error distributions (Fig. 2). To capture these heteroscedastic characteristics, our

method actively acquires repeated multi-view observations in a one-time offline initialization stage. In typical structured SDL environments, this initialization only needs to be performed once per tag-camera configuration; the computed empirical covariance matrices act as highly accurate priors for subsequent online operations over extended periods.

All fusion operations are performed in the Lie algebra $\mathfrak{se}(3)$ tangent space—a minimal, redundancy-free linear parameterization of $SE(3)$ —where small pose deviations can be linearly combined, enabling direct information-matrix-weighted averaging of pose residuals.

Initialization Stage (1) *Coarse localization*: Perform an initial single-view pose estimate to locate the fiducial ${}^b_t\mathbf{T}_0$. (2) *Viewpoint planning*: Generate target camera poses $\left\{\hat{{}_g^b\mathbf{T}_m}\right\}_{m=1}^M$ around the tag. (3) *Repeated sampling*: Repeat image acquisitions N times across the M viewpoints. For each image, independently estimate the tag pose ${}^b_t\mathbf{T}_{n,m}$. This empirical sampling directly captures the coupled errors from visual detection, hand-eye calibration, and kinematic inaccuracies. (4) *Covariance estimation*: Map the pose estimates to Lie algebra $\mathfrak{se}(3)$ residuals to compute the per-viewpoint covariance Σ_m and corresponding information matrices $\Lambda_m = (\Sigma_m + \varepsilon \mathbf{I})^{-1}$.

Working Stage During live operation, the robot first performs a single-shot coarse localization to obtain an approximate tag pose ${}^b_t\mathbf{T}'_0$. Based on this coarse estimate, the pre-planned target viewpoints $\{{}_g^b\mathbf{T}_m\}$ are transformed via ${}_g^b\mathbf{T}'_m = {}^b_t\mathbf{T}'_0 {}^b_t\mathbf{T}_0^{-1} {}_g^b\mathbf{T}_m$ into the current base frame, thereby compensating for navigation-induced base offsets. This preserves the validity of the pre-computed information matrices $\{\Lambda_m\}$. The robot then acquires a single set of images from the adjusted viewpoints, computes M independent poses $\{{}_t^b\mathbf{T}'_m\}$, and fuses them via the pre-computed weights to obtain the final result ${}^b_t\mathbf{T}_{\text{fused}}$.

Below we detail acquisition pose selection and multi-view pose fusion in the initialization stage.

3.3 Acquisition Pose Selection

To systematically sample target viewpoints $\left\{\hat{{}_g^b\mathbf{T}_m}\right\}$, we design a uniform spatial grid defined in the local coordinate frame of the tag. Let $\hat{\mathbf{t}}_m$ be the translation of the camera relative to the tag origin, and $\hat{\mathbf{R}}_m$ be the relative rotation. We enforce two geometric constraints on all viewpoints:

(1) **Target Centering Constraint**: The camera’s principal axis (the Z -axis) must point directly at the tag center to minimize boundary distortions, i.e., $\hat{\mathbf{R}}_m[0\ 0\ 1]^\top = -\frac{\hat{\mathbf{t}}_m}{\|\hat{\mathbf{t}}_m\|_2}$.

(2) **Roll Angle Constraint**: The X -axis of the camera must remain orthogonal to the Y -axis of the tag, i.e., $[0\ 1\ 0] \cdot \hat{\mathbf{R}}_m \cdot [1\ 0\ 0]^\top = 0$. This fixes the remaining roll degree of freedom, ensuring a consistent up-direction for the

target across all image captures. This constraint reduces random appearance variations, allowing for more consistent keypoint detection and reliable cross-viewpoint statistical modeling.

3.4 Multi-View Pose Fusion

Due to different viewing geometries across viewpoints, pose estimation error distributions vary. To fully exploit this variation, we construct information matrix weights for different viewpoint coordinate axes based on residual statistical properties in Lie algebra space, implementing multi-view pose weighted fusion.

This module's input is repeated multi-view pose estimates ${}^b_t\mathbf{T}_{n,m}$ where n denotes cycle and m denotes viewpoint. Output is the fused pose estimate ${}^b_t\mathbf{T}_{\text{fused}}$.

To simplify modeling and data acquisition, we do not explicitly estimate cross-viewpoint statistical correlations, approximating the joint observation noise covariance in block-diagonal form, modeling only the correlations among the 6 pose components within each viewpoint. This assumption typically holds when viewpoint images are acquired from different poses with independent single-view estimation.

Residual definition and information matrix estimation To perform linear weighting on SE(3) poses, we work in the $\mathfrak{se}(3)$ tangent space. Given pose ${}^b_t\mathbf{T}_{n,m}$ from cycle n , viewpoint m , we define the 6D residual vector using the logarithmic map:

$$\boldsymbol{\xi}_{n,m} = \log\left(\mathbf{T}_{\text{ref}}^{-1} {}^b_t\mathbf{T}_{n,m}\right)^\vee \in \mathbb{R}^6, \quad (2)$$

where $\mathbf{T}_{\text{ref}}$ is computed once during the offline initialization stage and remains fixed for all subsequent online operations. It averages the translation components and uses Markley's method [14] for the rotation components. For a fixed viewpoint m , we compute the sample mean $\bar{\boldsymbol{\xi}}_m$ and covariance Σ_m from N iterations. The information matrix is then defined as $\Lambda_m = (\Sigma_m + \varepsilon \mathbf{I})^{-1}$, where $\varepsilon = 10^{-12}$ prevents ill-conditioned Σ_m from becoming numerically singular, while being orders of magnitude smaller than typical diagonal entries of Σ_m to avoid biasing the estimates.

Multi-view weighted fusion During online fusion, for each viewpoint m 's current observation pose ${}^b_t\mathbf{T}_m$, the residual vector is constructed as:

$$\boldsymbol{\xi}_m = \log\left(\mathbf{T}_{\text{ref}}^{-1} {}^b_t\mathbf{T}_m\right)^\vee. \quad (3)$$

The residuals across viewpoints are weighted by the information matrices to obtain the fused residual:

$$\boldsymbol{\xi}_{\text{fused}} = \left(\sum_{m=1}^M \Lambda_m\right)^{-1} \left(\sum_{m=1}^M \Lambda_m \boldsymbol{\xi}_m\right). \quad (4)$$

Finally, the fused residual is mapped back to $\text{SE}(3)$ and combined with the reference pose to obtain the final fused pose:

$${}^b_t\mathbf{T}_{\text{fused}} = \mathbf{T}_{\text{ref}} \cdot \exp(\hat{\boldsymbol{\xi}}_{\text{fused}}), \quad (5)$$

where $\exp(\cdot)$ denotes the exponential map and $\wedge$ maps $\mathbb{R}^6$ to $\mathfrak{se}(3)$.

4 Experiments

4.1 Experimental Setup

We collect 120 independent multi-view acquisitions across 25 candidate viewpoints. The first $N = 60$ samples form the initialization set to compute the per-viewpoint information matrices, while the remaining 60 samples serve as the test set to evaluate repeatability.

Evaluation Metrics We evaluate *repeatability* (i.e., consistency across repeated localizations) rather than absolute task-space accuracy, as it is strictly required by the teach-and-replay paradigm and directly reflects system robustness against viewpoint-dependent noise. Three metrics are reported: RMSE_0 (translation RMSE on the tag origin), Angular RMSE (rotational RMSE of the estimated pose, denoted as Angle), and RMSE_{400} (translation RMSE evaluated at virtual target points $\mathbf{p} = [\pm 400, \pm 400, 0]^\top$ mm in the tag frame, which captures both translation and rotation-induced lever-arm errors). Translation errors are reported in μm and angular errors in μrad .

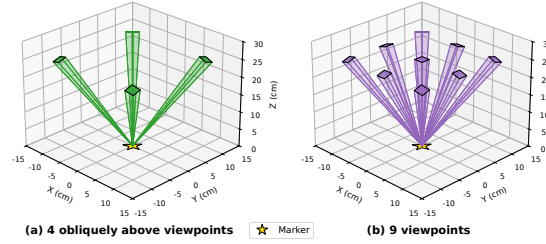


Fig. 3: Selected multi-view poses for fusion: (a) 4-view, and (b) 9-view configurations at $z = 25$ cm with target-centering.

Pose Selection The tag coordinate system is established with the center of the positioning tag as the origin, where the x -axis and y -axis are located in the tag plane, and the z -axis is perpendicular to the tag plane and points towards the camera side. We construct a regular translation grid in the tag coordinate

system and enforce consistent target-centering and roll constraints (as modeled in Section 3.3) for all viewpoints.

We use a 5×5 translation grid at a fixed height ($z = 25$ cm) as candidate viewpoints and evaluate representative subsets. Fig. 3 shows the directly-above, obliquely-above, 4-view and 9-view sets.

4.2 Comparison between Baseline Methods and the Weighted Fusion Method

Equal-weight Averaging This baseline directly averages multi-view poses without distinguishing viewpoint quality. We implement it as translation mean plus Markley quaternion averaging [14].

Minimizing Reprojection Error This baseline jointly optimizes the tag pose by minimizing the summed squared reprojection residuals across all views [2]. It can leverage multi-view redundancy, but may be sensitive to outliers and relies on simplified noise assumptions, making it less stable when viewpoint qualities vary.

Table 1: Comparison of multi-view fusion methods (units: RMSE in μm , angle in μrad).

Viewpoints	Method	RMSE ₀	Angle	RMSE ₄₀₀
4 obliquely above	Equal-weight Avg.	21.87	120.03	41.44
	Min. Reproj. Error [2]	13.25	97.91	31.63
	Weighted Fusion (Ours)	10.85	79.32	25.77
9 views	Equal-weight Avg.	24.22	164.89	52.95
	Min. Reproj. Error [2]	14.62	85.19	28.91
	Weighted Fusion (Ours)	13.91	74.91	25.95

Table 1 shows that our uncertainty-aware fusion outperforms both equal-weight averaging and reprojection minimization, reducing angular error by up to 19% over reprojection minimization and achieving the largest gains on RMSE₄₀₀—the metric most sensitive to small orientation errors. Notably, equal-weight averaging degrades when low-quality viewpoints are included (e.g., the 9-view set), confirming the necessity of information matrix weighting.

4.3 Precision of Single-View Pose Estimation Under Different Viewpoints

Fig. 4 shows the single-view RMSE₄₀₀ distribution across 25 candidate viewpoints ($x, y \in \{-10, -5, 0, 5, 10\}$ cm, $z = 25$ cm). The error exhibits strong spatial heterogeneity: the directly-above observation ($x=0, y=0$) reaches 325.9 μm , while obliquely-above views range from 51.1 to 130 μm —a $6.4\times$ difference.

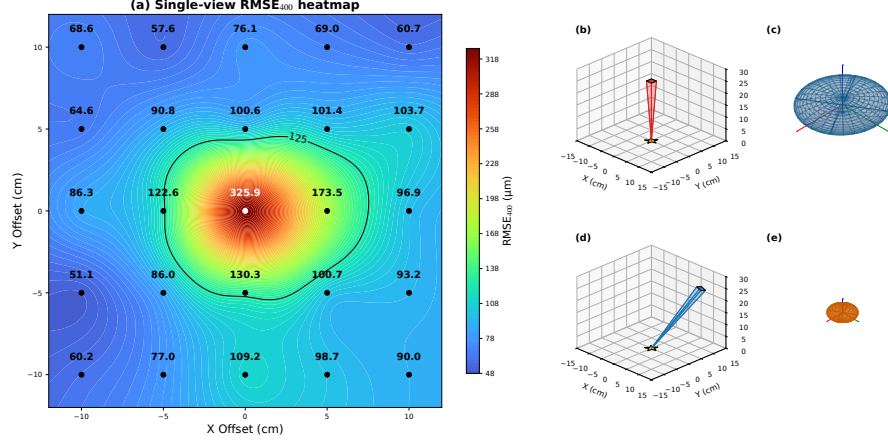


Fig. 4: Single-view localization quality. (a) RMSE₄₀₀ heatmap on a 5×5 grid with a sharp central peak ($325.9 \mu\text{m}$). (b,d) Directly-above and obliquely-above camera views. (c,e) Rotation uncertainty ellipsoids (sqrt-scale); the directly-above ellipsoid is elongated along the x and y axes, indicating inaccurate rotation estimation around these axes.

This reflects viewpoint-dependent geometric degradation: near-normal views reduce the tag’s apparent perspective distortion, amplifying depth and rotation uncertainties from corner detection noise; at $d = 400 \text{ mm}$, even small rotation errors are magnified into large position offsets. Fig. 4(c,e) confirms that the directly-above viewpoint yields an anisotropic, elongated uncertainty ellipsoid, whereas obliquely-above viewpoints produce compact ones. These per-viewpoint differences directly motivate information-matrix-weighted fusion.

Table 2: Comparison of single-view and multi-view fusion methods (units: RMSE in μm , angle in μrad).

Viewpoints	Method	RMSE ₀	Angle	RMSE ₄₀₀
1 directly above	Single-view	27.45	1147.30	325.94
1 obliquely above	Single-view	30.91	176.72	60.25
4 obliquely above	Weighted Fusion (Ours)	10.85	79.32	25.77

Table 2 highlights the difference between single-view estimation and our multi-view fusion approach. It demonstrates that (i) single-view performance varies substantially across viewpoints, with the directly-above observation exhibiting a remarkably high angular error ($1147.30 \mu\text{rad}$) and RMSE₄₀₀ ($325.94 \mu\text{m}$) due to geometric degeneracy, while the obliquely-above observation yields much

higher precision; and (ii) our multi-view weighted fusion consistently improves repeatability, surpassing even the best single-view estimate by reducing the angular error by 55% (from 176.72 to 79.32 μrad) and RMSE_0 by nearly 65%.

4.4 Ablation Study

Ablation: Weighting only with Diagonal Variances To assess the contribution of inter-component correlations, we compare two strategies under identical data:

- **Variance-Based Fusion:** Retains only the diagonal elements of each covariance matrix, i.e., $\Sigma'_m = \text{diag}(\Sigma_m)$, $A'_m = (\Sigma'_m + \varepsilon \mathbf{I})^{-1}$, assigning independent weights per degree of freedom but ignoring coupling among the six residual components.
- **Covariance-Based Fusion (Ours):** Uses the full covariance matrix, capturing coupled error structures such as simultaneous degradation in specific axes and enabling more effective suppression during fusion.

Table 3 shows that full-covariance weighting consistently achieves the lowest RMSE_0 and RMSE_{400} under both 4-view and 9-view settings. Compared with diagonal-only weighting, modeling inter-component correlations provides additional gains, especially on RMSE_{400} , indicating that coupled error structures matter in practice.

Table 3: Comparison between weighted fusion based on variances and full covariance matrices (units: μm).

Viewpoints	Fusion Method	RMSE_0	RMSE_{400}
4 obliquely above	Variance-Based Fusion	20.79	39.23
	Covariance-Based Fusion (Ours)	10.85	25.77
9 views	Variance-Based Fusion	23.41	37.49
	Covariance-Based Fusion (Ours)	13.91	25.95

4.5 Real-World Validation with Dial Indicator Measurements

To validate the proposed method under realistic disturbances including navigation deviations, we conduct a dial indicator touch experiment in a real chemical laboratory environment. Three controlled conditions (Fig. 5) decouple different error sources:

1. **Group A:** The base remains stationary without visual localization; the manipulator repeatedly moves from the same initial pose to the same target touch pose, evaluating the intrinsic repeatability of the manipulator.

2. **Group B:** The base remains stationary, but visual localization is performed before each touch, evaluating the combined precision of the vision system and manipulator.
3. **Group C:** The mobile base is driven to dock at the target station before each touch, followed by visual localization, simulating the full operational workflow.

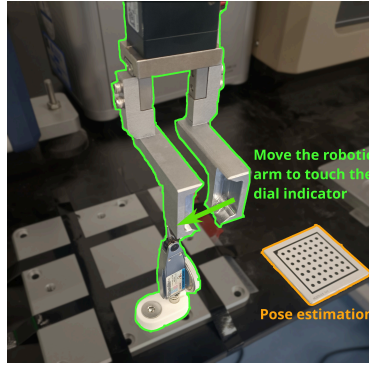


Fig. 5: Dial indicator touch experiment setup. After relocating the mobile base and performing pose estimation, the robot gripper contacts a dial indicator on the X-axis to measure positioning repeatability under different conditions.

Table 4: Statistical results of repeated dial indicator touch measurements under three conditions (unit: μm).

Experimental Conditions				Statistical Results			
Group	Move Base	Visual Loc.	Move Arm	n	s	90% CI for σ	95% PI for ΔX
A	No	No	Yes	14	5	[4, 8]	[-16, 16]
B	No	Yes	Yes	7	17	[12, 33]	[-59, 59]
C	Yes	Yes	Yes	6	23	[15, 43]	[-79, 79]

Table 4 summarizes the statistics. Group A (no vision, no base movement) shows excellent arm repeatability (95% PI $\pm 16 \mu\text{m}$). Group B (with vision) expands to $\pm 59 \mu\text{m}$, reflecting visual and hand-eye calibration uncertainty. Group C (full workflow with base movement) reaches $\pm 79 \mu\text{m}$, but the overall error remains within 0.1 mm, confirming sub-millimeter repeatability in real scenarios.

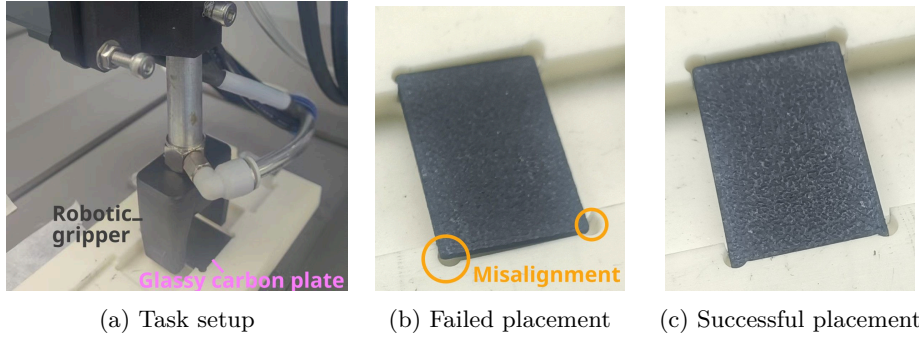


Fig. 6: Glassy carbon plate placement task. The robot places a carbon plate onto a workstation with a geometric tolerance of < 0.3 mm. (a) Task setup. (b) Failed placement (e.g., collision) due to baseline localization errors. (c) Successful placement using our multi-view localization.

4.6 Application to Chemical Laboratory Manipulation Tasks

To further validate the method in real scenarios, we apply the multi-view weighted fusion to a glassy carbon plate placement task (geometric tolerance < 0.3 mm). During setup, task poses are taught relative to the visual tag. At runtime, when the mobile base redocks with deviations, the fused pose ${}^b_t\mathbf{T}_{\text{fused}}$ converts the taught task poses into the current tag frame, enabling online correction without re-teaching. As shown in Fig. 6, the manipulator reliably places the plate across repeated trials, demonstrating that the proposed localization framework translates offline precision gains into operational reliability.

5 Conclusion

This study presents a high-precision multi-view pose estimation approach for mobile manipulators in self-driving laboratories. The key innovation is information matrix-weighted fusion in Lie algebra space, enabled by probabilistic uncertainty modeling during initialization. By utilizing repeated multi-view sampling to estimate residual covariances, our method adaptively prioritizes high-quality observations while suppressing unreliable, geometrically degraded viewpoints. Compared to conventional averaging or reprojection minimization, our approach improves repeatability. Ultimately, this framework improves the overall execution precision of mobile manipulators. The implementation of the proposed method will be made publicly available at <https://github.com/GWDx/multi-view-pose-estimation>.

Future work includes shrinkage-based precision estimation to reduce the bias of inverting sample covariances, extending uncertainty-aware weighting from the pose level to per-keypoint reprojection error minimization for end-to-end joint optimization, and developing full-chain uncertainty propagation models that de-

couple heterogeneous error sources to quantitatively guide hardware selection and calibration design.

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